

A Comparative Study of Random-Walk, Langevin, Barker, and Hamiltonian MCMC on PosteriorDB

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1 Introduction

Modern Bayesian workflows frequently require sampling from posterior distributions that are high-dimensional, strongly correlated, and often far from Gaussian. Markov chain Monte Carlo (MCMC) remains a principal tool for this task, but its practical success hinges on a trade-off: *efficiency* versus *robustness to tuning*. On the one hand, simple Metropolis–Hastings methods with symmetric random-walk proposals are reliable and broadly applicable, but they typically suffer from diffusive, random-walk exploration in moderate to high dimension. On the other hand, gradient-informed methods, such as the Metropolis-adjusted Langevin algorithm (MALA) and Hamiltonian Monte Carlo (HMC), can yield dramatically improved mixing by aligning proposals with the local geometry of the target, but this improvement comes at the cost of additional structure (gradients, integrator choices) and increased sensitivity to numerical parameters (step sizes, mass matrices, trajectory lengths).

This tension is amplified by the fact that “good” tuning is rarely a static, model-independent choice. Even for classical families of targets, scaling theory indicates that optimal efficiency occurs in narrow operating regions characterized by specific acceptance probabilities. For instance, diffusion-limit calculations suggest an optimal acceptance probability near 0.234 for the random-walk Metropolis algorithm on product targets [4], near 0.574 for MALA under similar asymptotics [5], and (in a separate asymptotic regime) near 0.651 for HMC when tuned via leapfrog step size [6]. These results are informative but also underscore a practical reality: the performance landscape can be steep, and departures from the tuned regime may cause rapid degradation.

An example is the *Barker proposal* of (author?) [7]. The method belongs to a broader class of *locally balanced* Metropolis–Hastings proposals, where the proposal mechanism itself is biased using first-order (gradient) information, but in a way designed to be less brittle than standard Langevin proposals. The Barker proposal is inspired by the classical Barker acceptance function [8] through a specific balancing choice, and it is motivated explicitly by the question: not only “how fast is the method when optimally tuned?”, but also “how badly does it behave when tuning is imperfect?” [7]. Closely related work shows that modifying the *auxiliary noise distribution* within the Barker construction can further improve performance, sometimes substantially, on representative targets [9].

The goal of this paper is therefore empirical and comparative: we implement a compact family of samplers—from adaptive random-walk Metropolis (AMH) to adaptive MALA, Barker-type proposals (with both Gaussian and bimodal auxiliary noise), and HMC—and benchmark them under a unified interface and diagnostic pipeline. The emphasis is methodological rather than purely computational: we report standard end-of-run convergence and efficiency diagnostics (e.g. $\hat{R}$, effective sample size, Monte Carlo standard error, runtime), and we also compute running-time performance curves that reveal *how quickly* each method approaches correct posterior moments. Crucially, we do not restrict attention to a single “best tuned” configuration. Instead, our framework is designed to evaluate both: (i) efficiency when adaptation succeeds, and (ii) robustness when tuning and geometry are more challenging.

2 Samplers

2.1 Common notation and the Metropolis–Hastings backbone

Let $\pi : \mathbb{R}^d \rightarrow (0, \infty)$ denote the target density in *unconstrained* coordinates, known up to a normalizing constant. (For Stan models, π is the posterior density after applying Stan’s unconstraining transform; see Section 4.) A generic Metropolis–Hastings (MH) transition from $x \in \mathbb{R}^d$ proceeds by:

1. proposing $y \sim q(\cdot \mid x)$, where $q(y \mid x)$ is a proposal density on $\mathbb{R}^d$,
2. accepting y with probability

$$\alpha_{\text{MH}}(x, y) = \min \left\{ 1, \frac{\pi(y) q(x \mid y)}{\pi(x) q(y \mid x)} \right\}, \quad (1)$$

and otherwise staying at x .

If the proposal is symmetric, $q(y | x) = q(x | y)$, then (1) reduces to $\alpha_{\text{MH}}(x, y) = \min\{1, \pi(y)/\pi(x)\}$ [10–12].

Across the samplers considered here, we separate *scale* and *shape*:

- A scale parameter $s > 0$ controls the overall magnitude of proposed moves.
- A shape matrix $\Sigma \in \mathbb{R}^{d \times d}$ (symmetric positive definite) controls anisotropy and correlation. We write $\Sigma = LL^\top$ with L lower triangular.

Our implementation exposes two families of shape adaptation: a diagonal (variance-only) estimate and a full (covariance) estimate, motivated by standard adaptive MCMC practice [1, 13].

2.2 Adaptive Metropolis–Hastings (AMH)

Idea. AMH is a random-walk Metropolis method whose proposal covariance is adapted during warm-up to match the empirical covariance of the target [1, 13]. It serves as a robust baseline: it requires only evaluations of $\log \pi(x)$ (no gradients), but it is vulnerable to random-walk scaling in high dimension [4].

Proposal. Given the current state $x_t \in \mathbb{R}^d$, propose

$$y = x_t + s_t L_t z, \quad z \sim \mathcal{N}(0, I_d), \quad L_t L_t^\top = \Sigma_t. \quad (2)$$

Equivalently, $y | x_t \sim \mathcal{N}(x_t, s_t^2 \Sigma_t)$, which is symmetric. The acceptance probability is therefore $\alpha_{\text{MH}}(x_t, y) = \min\{1, \pi(y)/\pi(x_t)\}$.

Adaptation (warm-up only). During warm-up, we update:

- *Scale adaptation:* we consider two alternatives in experiments: (i) a Robbins–Monro update targeting an acceptance probability $\tau^* \in (0, 1)$,

$$\log s_{t+1} = \log s_t + \gamma_t (\alpha_t - \tau^*), \quad \gamma_t = c t^{-\kappa}, \quad \kappa \in (1/2, 1], \quad (3)$$

and (ii) a dual-averaging update (also targeting τ^*), which typically produces smoother step-size trajectories during warm-up. In both cases α_t denotes the realized acceptance indicator (or acceptance probability) at iteration t .

- *Shape adaptation:* set Σ_t to either (i) a diagonal matrix of running marginal variances (cheap, ignores correlations) or (ii) a full empirical covariance estimate (captures correlations but is costlier). In both cases we use a regularization $\Sigma_t \leftarrow \Sigma_t + \epsilon I_d$ with small $\epsilon > 0$ to ensure positive definiteness.

After warm-up, (s_t, Σ_t) are frozen at their final values, yielding a homogeneous Markov chain for which standard MCMC diagnostics apply directly.

2.3 Adaptive Metropolis-adjusted Langevin algorithm (MALA)

Idea. MALA modifies the random walk by adding a drift term proportional to the gradient $\nabla \log \pi(x)$, approximating a Langevin diffusion whose stationary density is π [15]. The resulting proposal is asymmetric and must be corrected by MH. We refer to the full procedure with warm-up adaptation (scale and shape) as MALA; after warm-up, the resulting kernel is standard MALA with fixed tuning parameters.

Preconditioned MALA proposal. Given $x_t \in \mathbb{R}^d$, propose

$$y = x_t + \frac{s_t^2}{2} \Sigma_t \nabla \log \pi(x_t) + s_t L_t z, \quad z \sim \mathcal{N}(0, I_d), \quad L_t L_t^\top = \Sigma_t. \quad (4)$$

Thus

$$y | x_t \sim \mathcal{N}\left(x_t + \frac{s_t^2}{2} \Sigma_t \nabla \log \pi(x_t), s_t^2 \Sigma_t\right),$$

so $q(\cdot | x_t)$ is Gaussian but not symmetric.

MH correction. The acceptance probability is (1) with the MALA Gaussian proposal. Writing $m(x) = x + \frac{s^2}{2} \Sigma \nabla \log \pi(x)$ and using s, Σ fixed (post warm-up) for readability,

$$\log \frac{q(x | y)}{q(y | x)} = -\frac{1}{2s^2} \|L^{-1}(x - m(y))\|^2 + \frac{1}{2s^2} \|L^{-1}(y - m(x))\|^2, \quad (5)$$

and therefore

$$\alpha_{\text{MALA}}(x, y) = \min \left\{ 1, \exp \left(\log \pi(y) - \log \pi(x) + \log \frac{q(x | y)}{q(y | x)} \right) \right\}. \quad (6)$$

MALA can substantially outperform random-walk MH on smooth targets, but its performance is sensitive to step size. Under classical product-target asymptotics, the optimal acceptance probability is about 0.574 [5].

Adaptation. Our MALA implementation uses the same warm-up strategy as AMH: scale adaptation (Robbins–Monro or dual averaging) and either diagonal or full covariance shape adaptation. As with AMH, adaptation is disabled after warm-up.

2.4 The Barker proposal (locally balanced MH)

Idea. The Barker proposal of (author?) [7] is a gradient-based Metropolis–Hastings method constructed within a *locally balanced* framework. Rather than using the Langevin drift to shift the mean of a Gaussian proposal (as in MALA), the Barker proposal biases the *direction* of a symmetric base step using a logistic weighting derived from a *balancing function*. The resulting proposal is designed to maintain high efficiency while improving robustness to imperfect step-size tuning [7].

Balancing function and proposal kernel. Let $\mu_\sigma(\cdot)$ be a symmetric base density on $\mathbb{R}^d$ for increments (in practice Gaussian with scale and shape). A balancing function $g : (0, \infty) \rightarrow (0, \infty)$ is required to satisfy

$$g(t) = t g(1/t), \quad (7)$$

which ensures local reversibility properties in the locally balanced construction. The *Barker* choice is

$$g(t) = \frac{t}{1+t}. \quad (8)$$

With this choice, (author?) [7] show that the resulting continuous-state proposal density can be written in closed form as

$$q_B(y | x) = \frac{2 \mu_\sigma(y - x)}{1 + \exp\{-\langle \nabla \log \pi(x), (y - x) \rangle\}}, \quad x, y \in \mathbb{R}^d, \quad (9)$$

where the factor 2 arises from the normalization induced by the Barker balancing function.

Sampling mechanism (auxiliary-variable view). A key practical advantage is that one can sample from $q_B(\cdot | x)$ without needing to evaluate its normalizing constant: the method admits an auxiliary-variable construction (Algorithm 1 in [7]) that samples a base increment and then stochastically chooses a direction via logistic probabilities depending on $\nabla \log \pi(x)$. This is the mechanism implemented in the `rmcmc` package [16].

MH correction. Although inspired by Barker’s classical acceptance function [8], the Barker proposal here is used as an MH *proposal kernel* and is still corrected by the standard MH acceptance probability (1). In particular, because q_B is generally asymmetric, the acceptance probability is not $\pi(y)/(\pi(x) + \pi(y))$ (that expression applies to Barker’s accept/reject rule under symmetric proposals). For the Barker proposal, (author?) [7] derive a convenient form:

$$\alpha_B(x, y) = \min \left\{ 1, \frac{\pi(y) (1 + \exp\{\langle \nabla \log \pi(x), (y - x) \rangle\})}{\pi(x) (1 + \exp\{\langle \nabla \log \pi(y), (x - y) \rangle\})} \right\}. \quad (10)$$

The expression (10) is obtained by plugging (9) into (1) and exploiting symmetry of μ_σ .

Adaptation. We adapt scale and shape during warm-up exactly as in Sections 2.2–2.3, then freeze.

2.5 Bimodal Barker (noise-shape modification)

Idea. The Barker proposal depends on a choice of auxiliary “noise” distribution (equivalently, the base increment distribution μ_σ). (author?) [9] study the *optimal design* of the Barker proposal and related locally balanced MH algorithms, and they show that replacing Gaussian auxiliary noise by a carefully chosen *bimodal* noise distribution can improve performance (notably ESJD) on a range of targets.

Bimodal auxiliary noise used in our implementation. We follow the concrete choice implemented as `bimodal_barker_proposal()` in `rmcmc` [16]. Componentwise, the auxiliary noise distribution is an equally weighted mixture of two normals with shared variance σ^2 and means $\pm\sqrt{1-\sigma^2}$:

$$Z_i \sim \frac{1}{2} \mathcal{N}\left(+\sqrt{1-\sigma^2}, \sigma^2\right) + \frac{1}{2} \mathcal{N}\left(-\sqrt{1-\sigma^2}, \sigma^2\right), \quad i = 1, \dots, d, \quad (11)$$

so that $\mathbb{E}[Z_i] = 0$ and $\text{Var}(Z_i) = 1$ for each component. The scalar $\sigma \in (0, 1)$ controls how concentrated each mode is; the default in `rmcmc` is $\sigma = 0.1$ [16]. The Barker construction (proposal kernel plus MH correction) is then applied exactly as in Section 2.4, but with the auxiliary noise distribution replaced by (11).

2.6 Hamiltonian Monte Carlo (HMC): leapfrog dynamics, gradient cost, and Metropolis correction

Let $\pi(x)$ be the target density on $\mathbb{R}^d$ (known up to a normalizing constant). Define the *potential energy*

$$U(x) := -\log \pi(x), \quad \nabla U(x) = -\nabla \log \pi(x),$$

and introduce an auxiliary *momentum* variable $p \in \mathbb{R}^d$ with Gaussian distribution

$$p \sim \mathcal{N}(0, M),$$

where $M \in \mathbb{R}^{d \times d}$ is a symmetric positive definite *mass matrix*. The corresponding Hamiltonian is

$$H(x, p) := U(x) + \frac{1}{2} p^\top M^{-1} p.$$

HMC proposes distant moves by approximately simulating Hamiltonian dynamics using a symplectic integrator (the leapfrog method). Given the current position x_t , we proceed as follows.

Leapfrog integrator. Let $x^{(0)} = x_t$ and sample $p^{(0)} \sim \mathcal{N}(0, M)$. For $\ell = 0, 1, \dots, L-1$, perform one leapfrog step with step size $\epsilon > 0$:

$$p^{(\ell+\frac{1}{2})} = p^{(\ell)} - \frac{\epsilon}{2} \nabla U(x^{(\ell)}), \quad (12)$$

$$x^{(\ell+1)} = x^{(\ell)} + \epsilon M^{-1} p^{(\ell+\frac{1}{2})}, \quad (13)$$

$$p^{(\ell+1)} = p^{(\ell+\frac{1}{2})} - \frac{\epsilon}{2} \nabla U(x^{(\ell+1)}). \quad (14)$$

Denote the numerical endpoint by $(x^*, p^*) := (x^{(L)}, p^{(L)})$. In the standard reversible construction, one additionally negates the momentum, $p^* \leftarrow -p^*$ (this does not change the acceptance probability but ensures reversibility of the proposal mapping).

Gradient evaluation cost. Equations (12)–(14) require evaluating the gradient at

$$\nabla U(x^{(0)}), \nabla U(x^{(1)}), \dots, \nabla U(x^{(L)}),$$

i.e. typically $L+1$ gradient evaluations per HMC proposal trajectory (up to caching/reuse details).

Metropolis correction. Because leapfrog integration is not exact (finite ϵ), HMC applies a Metropolis correction targeting the joint density proportional to $\pi(x) \exp(-\frac{1}{2} p^\top M^{-1} p)$. The proposal is accepted with probability

$$\alpha((x_t, p^{(0)}), (x^*, p^*)) = \min\left\{1, \exp\left(-H(x^*, p^*) + H(x_t, p^{(0)})\right)\right\}.$$

Finally, set

$$x_{t+1} = \begin{cases} x^*, & \text{with probability } \alpha, \\ x_t, & \text{otherwise.} \end{cases}$$

Trajectory length and auxiliary updates (implementation interface). In our implementation, the step size ϵ corresponds to the `scale` parameter, while the choice of trajectory length is controlled either by a fixed L (`n_step`) or by randomizing L via a user-supplied function (`sample_n_step`). Additionally, the auxiliary-variable update `sample_auxiliary` can be used to implement alternatives to full momentum resampling, such as partial momentum refreshment.

3 Evaluation methodology

We evaluate samplers using both *end-of-run diagnostics* (convergence and efficiency summaries) and *running diagnostics* (how quickly posterior moments are learned). We typically run $M \geq 2$ chains per configuration, each with warm-up followed by N retained draws. In a small number of exploratory batches, we report single-chain results and interpret them as within-chain performance signals rather than definitive multi-chain convergence evidence. Let $\theta_t^{(m)}$ denote the (possibly vector-valued) draw at iteration t in chain m after warm-up.

3.1 Basic convergence and efficiency diagnostics

Split $\hat{R}$ (Gelman–Rubin). For a scalar estimand θ , define the per-chain means $\bar{\theta}_m$ and variances s_m^2 :

$$\bar{\theta}_m = \frac{1}{N} \sum_{t=1}^N \theta_t^{(m)}, \quad s_m^2 = \frac{1}{N-1} \sum_{t=1}^N \left(\theta_t^{(m)} - \bar{\theta}_m \right)^2.$$

Let $\bar{\theta} = \frac{1}{M} \sum_{m=1}^M \bar{\theta}_m$ be the overall mean. The within-chain variance and between-chain variance are

$$W = \frac{1}{M} \sum_{m=1}^M s_m^2, \quad B = \frac{N}{M-1} \sum_{m=1}^M (\bar{\theta}_m - \bar{\theta})^2.$$

An over-dispersed estimate of the marginal variance is

$$\hat{V} = \frac{N-1}{N} W + \frac{1}{N} B,$$

and the potential scale reduction factor is

$$\hat{R} = \sqrt{\frac{\hat{V}}{W}}. \quad (15)$$

Values near 1 indicate agreement between and within chains. In practice we use modern rank-normalized split- $\hat{R}$ variants as implemented in contemporary diagnostic toolchains [17].

Effective sample size (ESS). For correlated draws, ESS approximates the number of independent samples that would yield the same Monte Carlo accuracy. For a scalar estimand, a standard definition is

$$\text{ESS} = \frac{MN}{1 + 2 \sum_{k=1}^{\infty} \rho(k)}, \quad (16)$$

where $\rho(k)$ is the lag- k autocorrelation of the (appropriately combined) chain. In practice, the infinite sum is truncated using stable estimators (e.g. initial monotone sequence estimators) and computed separately for bulk and tail functionals [17]. We report:

- *Bulk-ESS*: ESS for rank-normalized draws, reflecting mixing in the central mass.
- *Tail-ESS*: ESS for extreme quantiles (e.g. 5% and 95%), reflecting tail exploration.

Monte Carlo standard error (MCSE). For an estimand $\psi = \mathbb{E}_{\pi}[h(\theta)]$ estimated by $\hat{\psi}$, the MCSE is

$$\text{MCSE}(\hat{\psi}) = \sqrt{\widehat{\text{Var}}(\hat{\psi})}.$$

For the posterior mean of a scalar parameter, a widely used approximation is

$$\text{MCSE}(\bar{\theta}) \approx \frac{\hat{\sigma}_{\theta}}{\sqrt{\text{ESS}}}, \quad (17)$$

where $\hat{\sigma}_{\theta}$ is the estimated marginal posterior standard deviation and ESS is as in (16). We compute MCSEs for means and selected quantiles.

Runtime and ESS/sec. We record wall-clock runtime T per sampler configuration, including gradient evaluations and any warm-up adaptation. To combine statistical and computational efficiency, we report

$$\text{ESS/sec} = \frac{\text{ESS}}{T}.$$

This normalization is essential when comparing gradient-free and gradient-based samplers, as their per-iteration costs can differ by orders of magnitude.

3.2 Running diagnostics (learning speed)

End-of-run diagnostics are necessary but do not capture *transient* behavior: two samplers might have similar final ESS yet converge to correct posterior moments at very different rates. When reference draws are available (PosteriorDB), we compute running-time metrics against “ground truth” moments.

Let $\theta \in \mathbb{R}^d$ denote a vector of parameters of interest in *constrained* coordinates (i.e. the model’s natural parameterization), and let (μ^*, v^*) be reference estimates of its mean and marginal variance vector computed from high-quality reference draws.

Running mean-squared error (MSE) of posterior means. Let $\hat{\mu}_n \in \mathbb{R}^d$ be the pooled running mean after n retained draws per chain (or equivalently after Mn pooled draws). Define

$$\text{MSE}_n = \frac{1}{d} \sum_{j=1}^d (\hat{\mu}_{n,j} - \mu_j^*)^2. \quad (18)$$

A fast sampler should drive MSE_n down quickly as n increases.

Running log-variance distance. Let $\hat{v}_n \in \mathbb{R}^d$ be the running estimate of marginal variances. We define the log-variance discrepancy

$$D_T(n) = \left(\frac{1}{d} \sum_{j=1}^d (\log \hat{v}_{n,j} - \log v_j^*)^2 \right)^{1/2}. \quad (19)$$

Using log stabilizes relative-scale errors: for example, over- and under-estimation by a factor of 2 are penalized equally because $|\log(2)| = |\log(1/2)|$.

Expected squared jump distance (ESJD). Let $x_t \in \mathbb{R}^d$ denote the unconstrained chain state (the coordinate system in which the Markov kernel operates). The ESJD over N retained draws is

$$\text{ESJD} = \frac{1}{N-1} \sum_{t=1}^{N-1} \|x_{t+1} - x_t\|^2, \quad (20)$$

computed per chain and then averaged across chains. ESJD is an exploration proxy: larger jumps suggest faster movement through state space. However, ESJD must be interpreted together with acceptance probabilities and moment diagnostics, because excessively large proposed jumps may simply be rejected. Moreover, ESJD is coordinate-dependent; in this paper it is computed in Stan’s unconstrained parameterization, so ESJD magnitudes should be compared within a given model/parameterization rather than treated as an absolute scale across unrelated targets.

4 Benchmark models

4.1 PosteriorDB targets and unconstrained parameterization

We benchmark on posteriors drawn from PosteriorDB, a curated collection of model–data pairs with reference posterior draws intended for testing and comparing inference algorithms [3]. For Stan models, we evaluate $\log \pi(x)$ and $\nabla \log \pi(x)$ in unconstrained coordinates $x \in \mathbb{R}^d$ using BridgeStan [2]. When reporting moment-based running diagnostics, we map unconstrained draws back to constrained coordinates to align with the reference draws.

In this paper we focus on:

- *Hierarchical geometry under a well-behaved parameterization* (Eight Schools, non-centered).
- *Multimodality and mode trapping* (a low-dimensional Gaussian mixture target).
- *Strong posterior correlation in unconstrained space* (**sblrc-blr**).

These targets are complementary: the Eight Schools model probes adaptation stability and typical hierarchical coupling under a non-centered reparameterization, the mixture model probes global exploration, and **sblrc-blr** probes whether adaptation and gradient-based proposals remain stable and efficient under strong correlation.

4.2 Eight Schools: non-centered parameterization

The Eight Schools model is a canonical hierarchical Bayesian benchmark [18, 19]. The data are $y_i \in \mathbb{R}$ with known standard errors $\sigma_i > 0$ for $i = 1, \dots, 8$. The hierarchical model is:

$$y_i \mid \theta_i \sim \mathcal{N}(\theta_i, \sigma_i^2), \quad i = 1, \dots, 8, \quad (21)$$

$$\theta_i \mid \mu, \tau \sim \mathcal{N}(\mu, \tau^2), \quad i = 1, \dots, 8, \quad (22)$$

$$\tau \sim \text{Cauchy}^+(0, 5), \quad (23)$$

with $\mu \in \mathbb{R}$ given the (improper) flat prior in the standard Stan formulation (i.e. no explicit prior term for μ is added). This is exactly the model structure used in the PosteriorDB Stan implementation.

In this paper we benchmark the *non-centered* form, which introduces auxiliary standard normals η_i :

$$\eta_i \stackrel{\text{iid}}{\sim} \mathcal{N}(0, 1), \quad \theta_i = \mu + \tau \eta_i. \quad (24)$$

This reparameterization often reduces posterior curvature and weakens the strongest τ - $\theta_{1:8}$ coupling when the data are weakly informative, improving practical sampling behavior [19]. We therefore use the non-centered target as a controlled benchmark for comparing adaptation stability and the relative efficiency of gradient-free versus gradient-based proposals, without simultaneously confounding the comparison with severe funnel pathologies associated with the centered parameterization.

4.3 Low-dimensional Gaussian mixture: `low_dim_gauss_mix-low_dim_gauss_mix`

To stress global exploration, we include the PosteriorDB target `low_dim_gauss_mix-low_dim_gauss_mix`, described in PosteriorDB as a *two-dimensional Gaussian mixture model*. Here “two-dimensional” refers to the mixture structure of the underlying model, whereas the sampler operates on the model’s unconstrained parameter vector, whose dimension is $d = 5$ in the PosteriorDB benchmark listings. Multimodal targets are a known stress test for local gradient methods: gradients tend to pull proposals toward the nearest mode, and energy barriers can lead to mode trapping.

This model is particularly relevant for Barker-type proposals because:

- the Barker construction modifies local proposal directionality without relying on a deterministic drift as in MALA, and
- bimodal auxiliary noise (Section 2.5) is explicitly motivated as a mechanism that can improve exploration and ESJD on representative targets [9].

We therefore use the mixture model to probe whether robustness-oriented proposals yield practical benefits beyond unimodal or near-Gaussian regimes.

4.4 `sblrc-blr`: correlated posterior in unconstrained coordinates

We also include the PosteriorDB target `sblrc-blr`, which yields a posterior with nontrivial correlation structure in Stan’s unconstrained parameterization. This model is used to stress-test adaptation procedures (especially full-covariance shape learning) and to assess whether gradient-informed proposals (MALA and Barker-type kernels) translate local geometric information into tangible gains when posterior correlation is strong. In particular, `sblrc-blr` provides a setting where diagonal adaptation may be insufficient to capture posterior geometry, while full covariance adaptation can be unstable if the warm-up estimate is ill-conditioned or overly noisy.

Scope note

This document defines the samplers, the benchmark targets, and the diagnostic methodology. No numerical results are reported here; results and discussion are presented in later sections of the full paper.

4.5 A sampler-agnostic adaptive warm-up wrapper

We implement each sampler through a common interface: for any choice of *scale* $s > 0$ and *shape* matrix $\Sigma \in \mathbb{R}^{d \times d}$ (symmetric positive definite), the sampler induces a Markov transition kernel

$$K_{s,\Sigma}(x, dx'),$$

Algorithm 1 Generic adaptive warm-up wrapper for a family of MCMC kernels $\{K_{s,\Sigma}\}$.

Require: Initial state $x_0 \in \mathbb{R}^d$; warm-up length T_w ; sampling length T_m ; initial scale $s_0 > 0$ and shape $\Sigma_0 \succ 0$; target acceptance probability δ .

Require: A *scale adapter* `ScaleUpdate` (stochastic approximation or dual averaging) and a *shape adapter* `ShapeUpdate` (diagonal “variance” or full “covariance”).

- 1: Initialize any internal state required by `ScaleUpdate` (e.g. running averages / dual variables).
- 2: **for** $t = 0, 1, \dots, T_w - 1$ **do**
- 3: **(Sampler step)** Draw $(x_{t+1}, \alpha_t) \sim \text{Step}(K_{s_t, \Sigma_t}, x_t)$, where α_t is the acceptance probability used in that step.
- 4: **(Scale adaptation)** Update the scale using the observed acceptance:

$$s_{t+1} \leftarrow \text{ScaleUpdate}(s_t, \alpha_t, t; \delta).$$

- 5: **(Shape adaptation)** Update the shape using the new state:

$$\Sigma_{t+1} \leftarrow \text{ShapeUpdate}(\Sigma_t, x_{t+1}, t).$$

- 6: **end for**
 - 7: **Freeze** (s, Σ) after warm-up (either $(s, \Sigma) = (s_{T_w}, \Sigma_{T_w})$ or a smoothed variant recommended by the scale adapter).
 - 8: **for** $t = T_w, \dots, T_w + T_m - 1$ **do**
 - 9: Draw $x_{t+1} \sim K_{s, \Sigma}(x_t, \cdot)$ (*no further adaptation*).
 - 10: **end for**
 - 11: **return** $\{x_t\}_{t=T_w}^{T_w+T_m}$.
-

which leaves π invariant when (s, Σ) are held fixed.¹ During an initial warm-up of T_w iterations we adapt (s, Σ) online, then freeze them and sample from the resulting fixed kernel.

Notation. Let $\pi(x)$ be the target density on $\mathbb{R}^d$ (known up to normalizing constant), and define $\ell(x) := \log \pi(x)$. When needed, we write $g(x) := \nabla \ell(x)$. Let $\delta \in (0, 1)$ denote the target acceptance probability. Let $\alpha_t \in [0, 1]$ denote the acceptance probability of the t -th proposal produced by K_{s_t, Σ_t} (for HMC, α_t is the Metropolis correction based on the Hamiltonian energy error). We consider two scale-adaptation mechanisms during warm-up: a Robbins–Monro stochastic approximation update and a dual-averaging update, both targeting δ .

Adaptive wrapper. The warm-up phase produces a sequence $\{(x_t, s_t, \Sigma_t)\}_{t=0}^{T_w}$ by alternating: (i) one step of the sampler at current (s_t, Σ_t) , (ii) an update of the scale s_t driven by α_t , and (iii) an update of the shape Σ_t driven by the realized states.

Role of the sampler-specific definitions. Algorithm 1 is agnostic to the choice of sampler. The only sampler-dependent component is the routine `Step`($K_{s,\Sigma}, x$), which must (i) generate the next state according to the selected kernel and (ii) report the associated acceptance probability α used by that kernel. In subsequent sections we specify $K_{s,\Sigma}$ and α for each sampler considered (AMH, MALA, Barker-type proposals, and HMC).

5 Benchmark results

5.1 Eight Schools (non-centered): comparative sampler behavior

We first benchmark all methods on the non-centered Eight Schools posterior `eight_schools-eight_schools_noncentered` (Section 4.2). This target serves as a “sanity” model: the non-centered reparameterization largely removes the funnel coupling, so differences between samplers are driven primarily by (i) the effectiveness and stability of warm-up adaptation, (ii) the ability to learn marginal scales, and (iii) computational cost. Performance is assessed using the running diagnostics defined in Section 3.2: running mean MSE (18), running log-variance distance $D_T(n)$ (19), and ESJD (20).

¹For HMC, Σ is interpreted as a mass matrix (or its inverse, depending on convention), whereas for Metropolis–Hastings proposals Σ is a proposal covariance/preconditioner.

Adaptive random-walk baseline (AMH). Among AMH configurations, diagonal (variance-only) shape adaptation consistently provides the best performance on this target. Both diagonal AMH variants rapidly reach a low plateau in running MSE and attain the smallest $D_T(n)$ values, while maintaining substantially larger ESJD than the corresponding full-covariance AMH variants. In contrast, full covariance adaptation performs poorly in this experiment: the covariance-shaped AMH runs exhibit (i) persistently larger MSE and $D_T(n)$ and (ii) markedly reduced ESJD, indicating that accepted moves are small and exploration is hindered. Operationally, this pattern is consistent with an ill-conditioned or mis-scaled learned covariance (e.g. noisy or unstable full-cov estimates), which can collapse the effective proposal geometry even when regularization is applied. On this non-centered target, where posterior correlations are comparatively mild, the diagonal adaptation appears to be both sufficient and more robust.

Gradient-based Metropolis proposals (MALA, Barker, Bimodal Barker). For the non-centered Eight Schools posterior, MALA, Barker, and Bimodal Barker achieve similar end-state running MSE, suggesting that mean estimation is not a separating stressor once geometry is regularized by the parameterization. Differences are more visible in variance learning and exploration proxies: MALA and Bimodal Barker reach smaller $D_T(n)$ values than standard Barker, and their ESJD traces are larger, indicating more aggressive movement in unconstrained space. The bimodal auxiliary noise improves over standard Barker in this experiment, consistent with its design goal of increasing typical jump distances while retaining MH correction. We note, however, that these gradient-MH results were obtained with a single chain in this batch, so they should be interpreted as within-chain performance signals rather than definitive multi-chain convergence evidence.

Hamiltonian Monte Carlo (HMC): trajectory length sensitivity and cost. HMC performance is strongly sensitive to the trajectory length parameter L in this benchmark. With short trajectories (fixed $L = 10$), ESJD is extremely small and both running MSE and $D_T(n)$ decay more slowly, indicating that the method behaves like an overly conservative local sampler. Increasing the trajectory length (fixed $L = 50$) substantially improves exploration and accelerates convergence of both means and variances. An adaptive step-count variant with auxiliary refresh further increases ESJD, but delivers only marginal additional improvement in running MSE/ $D_T(n)$ on this already-regular target. Because each HMC proposal requires $\mathcal{O}(L)$ gradient evaluations, these gains must be interpreted together with runtime: on the non-centered Eight Schools posterior, longer trajectories increase statistical efficiency per iteration but can be dominated by gradient cost in ESS/sec comparisons. (In our implementation, HMC was run with conservative, stability-oriented settings; in particular, step-size adaptation was disabled when required to avoid gradient overflows.)

Summary and practical takeaways (this model). On the non-centered Eight Schools target, most samplers reach comparable accuracy in posterior means, so the most informative separation comes from variance learning (D_T), exploration (ESJD), and computational efficiency. Diagonal AMH provides a strong gradient-free baseline with stable adaptation and good moment learning. MALA and Bimodal Barker exhibit stronger exploration (higher ESJD) and good variance learning, while standard Barker is comparatively weaker in this batch. HMC requires sufficiently long (or randomized) trajectories to behave as an effective global sampler; however, on this target the additional compute from large L can yield limited practical benefit beyond the point where running MSE and D_T have already stabilized.

5.2 Low-dimensional Gaussian mixture (low_dim_gauss_mix): multimodality, mode trapping, and why ESJD/ESS can mislead

This target is explicitly *multimodal*. The running diagnostics in Section 3.2 are computed against PosteriorDB reference moments, i.e. the *global* mixture mean and variances. Consequently, any sampler that remains confined to a single mode will typically exhibit (i) a plateau in the running mean error MSE_n (18) and (ii) a plateau in the running log-variance discrepancy $D_T(n)$ (19), even if within-mode mixing is excellent.

AMH (four configurations): diagonal shape succeeds; full covariance can fail on multimodality. Comparing the four AMH variants (stochastic-approximation vs dual-averaging scale adaptation; diagonal “variance” vs full “covariance” shape), two robust patterns emerge:

1. **Diagonal shape (variance) learns the global moments quickly.** Both variance-shape runs drive MSE_n and $D_T(n)$ essentially to zero over the main sampling window, indicating that the empirical means and marginal variances match the reference moments.
2. **Full covariance shape can be actively harmful on a mixture.** Both covariance-shape runs plateau at substantially larger MSE_n and $D_T(n)$ (and also exhibit poorer end-of-run mixing diagnostics relative to the diagonal runs), indicating that the chains are not reproducing the global mixture moments. A

plausible mechanism is that covariance adaptation is *mode-local*: if warm-up samples concentrate in one mode, the empirical covariance largely reflects within-mode correlations and scales. The resulting proposal can become well-tuned for local exploration but poorly aligned with the inter-mode direction, reducing the probability of barrier-crossing moves needed for correct global averaging.

A key cautionary example is that a configuration can attain a large ESJD while still performing poorly in MSE_n and $D_T(n)$; on multimodal targets, this can reflect rapid *within-mode* movement rather than genuine *between-mode* exploration.

Gradient-based MH (MALA, Barker, Bimodal Barker): strong local movement, wrong global moments under single-chain initialization. Across MALA, Barker, and Bimodal Barker, ESJD values can be high and stable, consistent with efficient local exploration once warm-up has stabilized scale/shape. However, all three methods can exhibit non-vanishing plateaus in MSE_n and $D_T(n)$ (at values far from zero), which is the diagnostic signature of *mode trapping*: the chain(s) explore one mode thoroughly but do not traverse the low-density region separating modes often enough to recover the global mixture moments.

This behavior is expected for locally informed proposals on multimodal landscapes. In MALA, the drift term $\Sigma \nabla \log \pi(x)$ points toward increasing log density, i.e. back into the current basin. For Barker-type proposals, the logistic tilting in (9) similarly biases directions according to the local gradient. Such mechanisms improve robustness and efficiency *within* a mode, but they do not by themselves create a global transition mechanism for crossing energy barriers. The bimodal auxiliary noise is designed to improve jump behavior in locally balanced schemes, but in this particular run it did not materially reduce the global-moment error, suggesting that the remaining limitation is inter-mode communication rather than within-mode step length. We emphasize that these results are based on a single chain in this batch; they should therefore be read as within-chain performance signals rather than definitive multi-chain convergence evidence.

HMC (trajectory-length sensitivity): larger ESJD with longer/adaptive paths, but still global-moment plateaus. As expected, increasing trajectory length (and/or randomizing it) can substantially increase ESJD and improve *local* exploration. Crucially, however, HMC can still fail to drive MSE_n or $D_T(n)$ toward zero on multimodal targets: both diagnostics can plateau well above zero. This is consistent with a standard multimodality failure mode for HMC: Hamiltonian trajectories approximately conserve total energy, so moving between modes typically requires traversing low-density regions corresponding to high potential energy. Longer trajectories improve exploration *within* a basin, but do not guarantee inter-basin transitions unless the momentum/step-size regime yields barrier crossing with non-negligible probability.

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